Lesson 5: Dendroarchaeology — handout

Reading Environment and History through Tree Rings | 90 minutes | Draft v1 | 30 October 2026

Central question

What can a calendar date assigned to wood tell us about human activity? Distinguish the outer-ring date, felling date and construction date. Explain conclusions and limits using preservation and archaeological context.

Interpretation sequence

1. Identify species, measurement path and preservation; crossdate the ring series.

2. Establish whether the last preserved ring is the original outermost ring.

3. Record bark edge, sapwood and the heartwood–sapwood boundary.

4. Separate final growth, felling, working, storage, construction and use.

5. Interpret dates with member positions, joints, tool marks, repairs and deposition.

Qualifications

Even a complete bark edge may belong to a tree that died before it was cut. A completed outer ring can indicate the following dormant interval, which may span calendar years. Seasonal interpretations require anatomical observations and local growth phenology.

A terminus post quem (TPQ) excludes earlier dates; it is not a precise construction date. Unknown loss of outer rings leaves the upper bound unresolved. Sapwood estimates need an appropriate species and regional reference population: there is no universal number to add.

Minimum record

Include specimen ID, structural location and relationships, species, measurement path, last ring, bark and sapwood preservation, crossdating evidence, meaning of the date, assumptions and alternatives. Record undated members too. Multiple members can come from one tree and may not provide independent confirmation.

Exercise A: constraining felling dates

8 minutes: individual work 4; pairs 3; record conclusions 1.

All cases below are fictional teaching examples, not empirical data.

ID	Observations and assumptions	Date information
A	Complete bark edge and an incomplete outer ring. Assume the tree was alive when cut in the observed growth season; local phenology is known.	Outermost ring: 1600
B	Heartwood only. Original outer edge and amount of loss unknown.	Last preserved ring: 1580
C	Heartwood–sapwood boundary identified; five sapwood rings survive. Assume 15–25 sapwood rings in total.	Last heartwood ring: 1590; preserved sapwood: 1591–1595

For each specimen

1. State what is established, estimated and unknown about felling.

2. Can construction be dated? Give a reason.

3. Suggest one further observation that would improve the interpretation.

Calculation for C

Last heartwood year + estimated total sapwood count = ( ) to ( ).

Or: last surviving sapwood year + estimated missing sapwood count = ( ) to ( ).

The 15–25 range is a teaching assumption, not a validated model for Japanese timber or a confidence interval established from real observations. Treat the estimated last growth year as an approximate felling year for this calculation; seasonal uncertainty, including a dormant interval crossing calendar years, is a separate issue.

Exercise B: construction, repair and reuse

8 minutes: interpret 3; compare alternatives 3; record 2. All examples are fictional.

ID	Position and observations	Date information
P1–P3	Assume original main-frame members. Complete bark edges; trees cut alive in their dated growth seasons.	Outermost rings: 1600, 1601, 1601
R1–R2	Roof members with complete bark edges; same living-tree and growth-season assumptions.	Both outermost rings: 1640
X	Beam; heartwood only; outer loss unknown. A redundant joint is present.	Last retained ring: 1530

Questions

1. What is the strongest lower date bound for the sampled main-frame arrangement?

2. Propose one interpretation of the younger roof timber.

3. Is X certainly reused? Give another explanation as well.

4. Choose an additional observation to compare the alternatives.

Separate levels of evidence

Write a short statement distinguishing observations, conditional inferences and unresolved questions. A cluster of dates does not exclude storage or same-tree origin. If the original-member assumption fails, revise the main-frame inference.

Compare published reports

Mitsutani (2001, pp.32–33) reports a 594 CE ring date for the Horyuji central pillar, discussing outer-wood preservation and three additional outer rings. Does a single member date establish the date of the entire building? See R31 for the original images and graph.

The Rook Hall Farm report (R32) derives winter 1522/23 felling from complete sapwood in one sample and suggests construction in 1523 or within a year or two. This is a contextual interpretation, not a universal storage allowance.

Terms and further study

Term	Brief explanation
dendroarchaeology	Using tree rings to study archaeological material and history.
last preserved ring	Youngest surviving ring; not necessarily the original outermost ring.
bark edge	Original outer wood surface directly beneath bark.
sapwood	Outer wood outside the heartwood; ring count varies.
heartwood–sapwood boundary	Boundary used as the origin for total-sapwood estimates.
terminus post quem (TPQ)	Earliest bound: an event cannot be earlier than this.
felling date	Time of cutting, distinguished from ring formation.
building phase	A proposed stage of construction, extension or repair.
reused timber	Timber used again after an earlier use.
dendroprovenancing	Investigating timber origin using ring patterns and other evidence.

Primary reports and database

- R30 Interpreting tree-ring dates — Historic England
- R31 Horyuji central-pillar dating report — 奈良文化財研究所
- R32 Rook Hall Farm timber analysis — Historic England
- R33 Building dendrochronology database — Archaeology Data Service
- R34 Izusan Gongen statue: timber and production dates — 奈良国立博物館

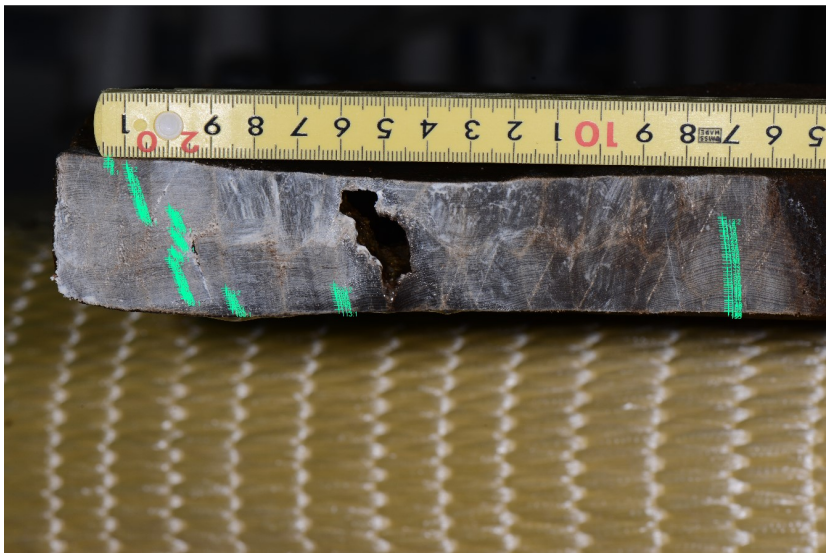
Link to the final assignment

Develop a research plan or review that identifies the question, material, crossdating evidence, preservation, interpretation assumptions and alternatives. Consult the syllabus requirement of 2–4 A4 pages and at least three references. Next: dendroecology.

Review: write approximately 200 Japanese characters (or a comparable short English summary) about a conclusion and its limits, add one reference and update the research outline.

Lesson 5 Trace evidence through a specimen ID

About eight minutes, replacing one case-reading activity. Use the printed evidence below; web exploration can wait until revision.



Nabunken specimen 29ho018. Green marks are published measurement locations. The entire image, including ruler and annotations, is retained.

Nara National Research Institute for Cultural Properties, 2024. CC BY 4.0.

Field	Information in the public record
ID and object	29ho018: lower well-frame board A1, well SE9650
Species	Hinoki cypress (Chamaecyparis obtusa)
Images	YH1_9812.JPG; annotated YH1_9812_29ho018.jpg
Width series	29ho018.csv: 155 rings, inner to outer, in 0.01 mm
First three values	32, 44, 42 = 0.32, 0.44, 0.42 mm
Not supplied in this CSV	Calendar years, reference chronology, match statistics, felling date

Connect the ID, image and width series. Correct the claim that 155 rings means the timber is 155 years old. What additional record is needed to assess its dating?

Nabunken 29ho018, CC BY 4.0

Lesson 5 Public data and the basis of interpretation

Check

155 is the number of measured rings, not elapsed years before the present. The CSV has no calendar-year column. Images and widths support checking the measurements. Assessing the date also requires the reference chronology, overlap, match results and their evaluation.

Follow the dataset description, metadata, specimen image and CSV, then the relevant dendrochronology section of Monograph 103. Record the reported date, outermost condition and basis of crossdating. Mark an unlocated item as unverified rather than filling it by inference.

Claim	Additional evidence
Calendar year of each ring	Reference series, candidate alignments, overlap, visual and statistical checks
Felling date	Bark edge, sapwood and wood lost through working
Construction and use of the well	Excavation context, stratigraphy, other finds and reuse traces

If 29ho017 is used for comparison, does a separate board necessarily represent a separate tree?

Different boards could come from one tree. Different IDs alone do not establish biological independence.

Dataset entry

Nabunken Monograph 103: excavation report